

We denote B_G the set of chief blocks of a group G

Def: Suppose $\alpha \in B_G$. We say that the block $\alpha \in B_G$ is robust ~~block~~ if none of the representatives of α are compact or quasi-discrete. The collection of ~~robust~~ robust chief blocks is denoted by B_G^r .

Def: ((Robust) block property)

A property P of groups is a (robust) block property if for every l.c.s.c. group G and α a (robust) chief block of G , there exists $K/L \in \alpha$ with P if and only if every $K/L \in \alpha$ has P . For a (robust) block property P , we say a (robust) chief block α of G has property P if some, equivalently all, $K/L \in \alpha$ has property P .

Thm:

The following are block properties: Being elementary with rank $\leq \alpha$, being amenable, being quasi-discrete, and being iso to a given connected group H . Compact generation is a robust block property.

Coro:

Suppose G is a l.c.s.c. group and suppose that α is a chief block of G that is not robust. Then either every representative of α is isomorphic to a fixed connected compact group, or every representative of α is t.d. and quasi-discrete.

Def: (Covering a block)

For a Polish group G , a closed normal subgroup $N \trianglelefteq G$ covers a block $\alpha \in B_G$ if there is $K/L \in \alpha$ s.t. $K \leq N$. A block α is minimally covered if there is a least normal subgroup that covers the block.

def: (extension of blocks)

Let G be a Polish group and H be a closed subgroup of G . Given $a \in B_H$ and $b \in B_G$, we say that b is an extension of a in G if every closed normal subgroup K of G , the group K covers b iff $K \cap H$ covers a .

Thm:

If $\mathcal{Q}$ is a compactly generated open subgroup of an l.c.s.c. group G , then all robust blocks of $\mathcal{Q}$ extend to G .

def: (Regionally robust blocks)

Blocks on a l.c.s.c. group arising from extending ~~that~~ a robust block of a compactly generated open subgroup are called regionally robust blocks. The set of regionally robust blocks of an l.c.s.c. group G is denoted B_G^{rr} .